

Revised Generation Interconnection System Impact Study Report

For Queue Project Q23-2321

North Bay 230 kV
July 22, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q23-2321. The Interconnection Customer proposes a 213.3 MW wind facility interconnecting at the North Bay 230 kV substation in the Ohio area. The request was received on June 12, 2023. The study evaluated system intact and single contingency conditions using the regional planning case, including 616 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$18,140,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Timken Richville - Sunnyside Substation 230 KV	Portage Lakes - Weedpatch 230 KV	110.1	457	38.1
Sunnyside - N Bay 230 KV	Portage Lakes - Weedpatch 230 KV L	103.0	516	38.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Timken Richville - Sunnyside Substation 230 KV	Rebuild line with higher-capacity conductor	\$14,811,000	66% (\$9,820,000)
Sunnyside - N Bay 230 KV	Reconductor line	\$12,549,000	66% (\$8,320,000)
	Total allocated to project		\$18,140,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.